

GSoC ROADMAP

PHASE 1: FOUNDATION (Month 1)

1. Choose ONE programming language

2. Learn Git & GitHub

You must know:

- What is a repository
- Fork vs clone
- Commit & push
- Pull Request (PR)

3. Very basic DSA (Only fundamentals)

Learn:

- Arrays
- Strings
- Loops
- Conditions
- Functions

PHASE 2: UNDERSTANDING OPEN SOURCE (Month 2)

4. Explore GSoC organizations (Correct way)

1. Go to Google Summer of Code website
2. Open Organizations / Org Ideas
3. Pick 2–3 organizations only

How to shortlist:

- Tech you understand or want to learn
- Clear README & documentation
- Active GitHub (recent commits/issues)

5. Understand a project

For ONE project:

1. Read README slowly
2. Answer these:
 - What problem does this project solve?
 - Who uses it?
 - What does success look like?

PHASE 3: ISSUES & FIRST CONTRIBUTION (Month 3)

6. How to read issues properly

Go to project repo → Issues

Look for labels:

- good first issue
- beginner
- documentation
- easy

For every issue, understand:

1. What is wrong / missing?
2. Which file/module is involved?
3. What is expected after fix?

7. Run the project locally (Very important)

Steps:

1. Fork repo
2. Clone it
3. Follow README to run locally
4. Observe:
 - How project starts
 - What happens when you change code

8. Make your first contribution

Start with:

- Documentation improvement
- Typo fix
- Small bug
- Simple logic change

Steps:

1. Create a new branch
2. Make small change
3. Commit clearly
4. Open Pull Request

PHASE 4: COMMUNICATION & PROPOSAL (Month 4)

9. Writing the GSoC proposal (Heart of selection)

Proposal should include:

1. Abstract (short summary)
2. Problem statement
3. Your solution approach
4. Weekly timeline
5. Benefits to organization
6. Your past contributions

PHASE 5: CODING PERIOD (After Selection)

- Follow weekly milestones
- Communicate blockers
- Improve code quality
- Learn real-world dev practices